

Adapt Without Losing the Reader: A Modular Two-Screen Platform for Gaze-Contingent Reading Experiments

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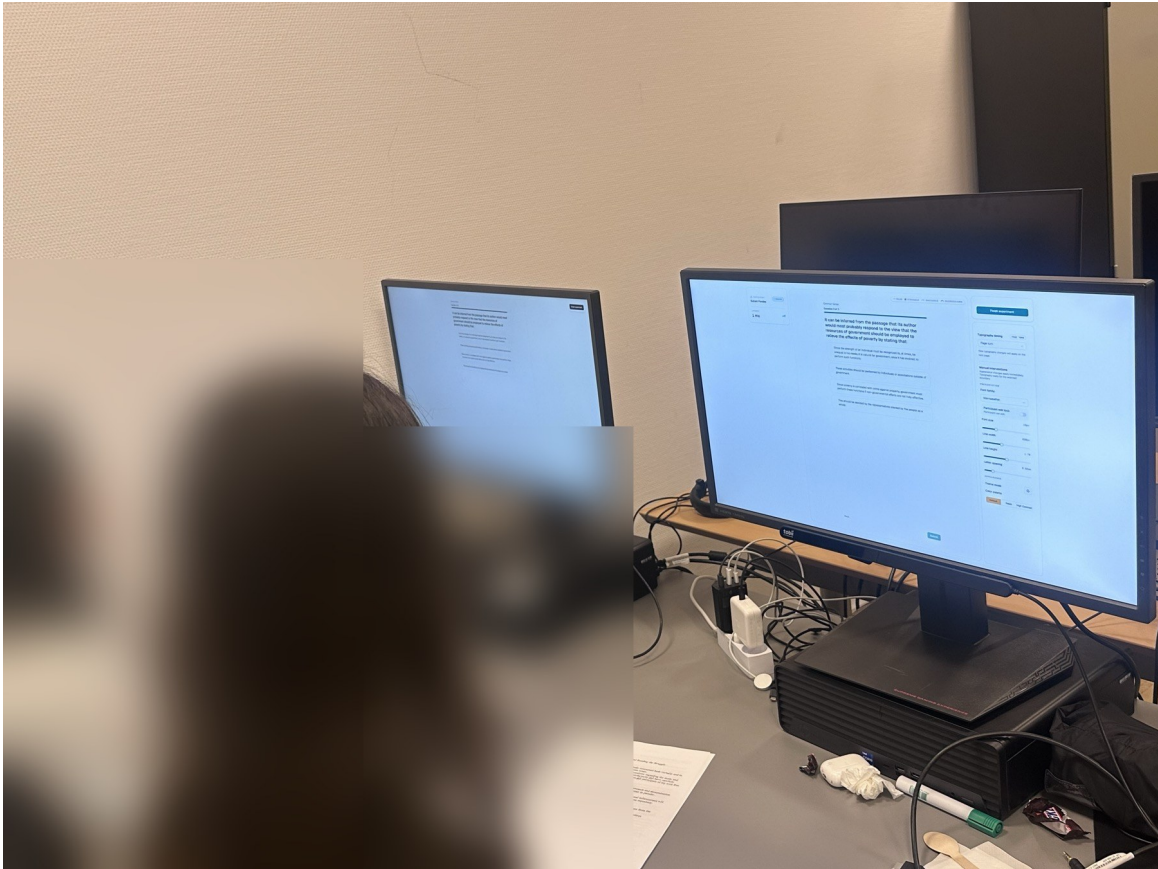


Figure 1: The two-screen setup during an evaluation session. The participant reads the presented text on one display (left), while the researcher monitors the mirrored view and applies interventions from the console on the other (right); the eye tracker is mounted at the foot of the participant’s display, and both surfaces are served by one backend on a single host. Faces are blurred for anonymity.

Abstract

Real-time gaze-driven text adaptation has been demonstrated repeatedly, but existing prototypes couple sensing, decision logic, and the reading interface, so studying which interventions actually help remains expensive. We present a researcher-operated, two-screen platform in which sensing, eye-movement analysis, decision strategy, and intervention execution are independently replaceable modules: a participant reads reflowable text that adapts to their gaze, while a researcher mirrors the reading live and applies or approves typographic interventions, with every gaze sample, decision, and intervention captured in a replayable session record.

Across eight reading sessions on real eye-tracking hardware, the platform sustained the tracker’s rated 90 Hz with gaze validity above 92%, kept every measured latency sample within its 100 ms budget, and had its evaluation sessions analysed live by an independent team’s reading-difficulty pipeline connected through the platform’s provider contract without any change to the core. Because the platform measures the positional cost of each intervention it applies, it revealed that our original context-preserving restore moved the reader about three times further than the reflow it compensated; we corrected the mechanism and verified the correction over a 66-trial deterministic sweep. The platform turns comparisons of

reading interventions and decision strategies, including AI-driven ones, into configuration rather than reimplementing.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; *Empirical studies in HCI*; *Accessibility systems and tools*.

Keywords

eye tracking, adaptive reading, gaze-contingent interfaces, typographic interventions, context preservation, research platforms

1 Introduction

Reading underpins participation in education, employment, and daily life, yet for many people it is far from effortless. The World Health Organization estimates that at least 2.2 billion people live with a vision impairment [24], age-related macular degeneration alone is projected to affect 288 million people by 2040 [1], and reading difficulties such as dyslexia add to the population for whom text demands more than the visual or cognitive system can comfortably supply. Digital text offers a response that print cannot: its presentation can change while it is being read. Font size, spacing, line length, and contrast are all adjustable on screen, controlled readability studies show that such parameters measurably affect reading [2, 3, 20], and an eye tracker supplies the moment-to-moment signal needed to apply an adjustment exactly when a reader needs it. A system that couples the two can deliver individualised typographic micro-interventions [17] without the reader ever issuing a command.

The promise carries a structural risk. A typographic change that helps in principle also reflows the document, so the line a participant was reading moves, and an ill-timed change costs the reader their place in the very flow the system is trying to protect. Prior work confirms both sides of this tension: gaze-driven typographic adaptation is technically achievable, and the design of the intervention, in particular how the transition is presented and whether the reader's context is preserved, measurably shapes recovery and acceptance [8, 16].

We argue that the bottleneck in this research area has shifted. Demonstrating that gaze-driven adaptation works no longer requires another prototype; three generations of adaptive readers in the Reading the Reader programme alone have shown it [14, 16, 19]. What is missing is the instrument for studying it: each of those prototypes wove sensing, decision logic, and the reading interface into one codebase, so each new study extended the previous system by forking it, and none offered a researcher console for operating a controlled session or a session record from which an experiment could be reproduced. Consider a researcher who wants to test whether fading a font change gradually into place disrupts a reader less than applying it abruptly. On a coupled prototype this scientifically simple comparison means editing core application code and maintaining two divergent forks under identical sensing. An experiment that should be a configuration choice is instead a software project.

This paper presents a platform built to remove that friction, shown in Figure 1. A participant reads reflowable text on one screen while the text adapts to their gaze; on a second screen, a researcher

follows the participant's gaze, fixations, and regressions live over a mirror of the same text, and applies typographic interventions or approves machine-proposed ones during the session. Behind this workflow, the adaptive loop is decomposed into four independently replaceable modules (sensing, eye-movement analysis, decision strategy, and intervention execution) connected through stable contracts, and every gaze sample, decision, and intervention is written into one schema-versioned session record that can be exported, replayed, and re-analysed.

The contributions are:

- **A modular platform for gaze-contingent reading experiments**, in which sensing, analysis, decision, and intervention are separate modules behind stable contracts, extendable in process by registration and out of process through a documented provider protocol with graceful fallback.
- **A two-screen, researcher-in-the-loop workflow**, in which interventions are applied manually, proposed by a decision provider and gated by the researcher (advisory), or applied autonomously, without structural change between modes.
- **A context-preserving intervention mechanism that measures its own cost**: layout changes commit at controlled boundaries, a continuously captured anchor restores the reader's position across the reflow, and a graded positional outcome for every intervention is written into the session record.
- **An evaluation on real eye-tracking hardware** covering real-time feasibility (rated 90 Hz sensing, every measured latency sample within a 100 ms budget), external-provider integration (three providers, including an independent team's reading-difficulty pipeline that analysed the evaluation sessions live), and context preservation (a with-and-without comparison, and a 66-trial deterministic sweep). The platform's self-measurement exposed a defect in our original restore strategy, which over-repositioned the reader on small reflows; we describe the revision and its geometric verification.

The platform, its documentation, and the analysis pipeline behind every figure and table in this paper are openly available.

2 Related Work

2.1 Gaze-Contingent and Adaptive Reading

Using the reader's gaze to change the reading experience in real time has a long line of demonstrations. The eyeBook coupled an eye tracker to a digital book to trigger in-situ effects as the reader progressed [4]. More recently, Medan and Pelman's New Line interface presented text as a single line that scrolls with the reader's webcam-estimated gaze and emphasises the fixated word; a pilot with 13 participants found no significant change in comprehension or speed but a clear preference for the interface [10]. Rummens and Beier coloured the word under fixation in real time for seventy second graders reading aloud and reported faster reading with shorter fixations, fewer regressions, and less re-reading [21]. Gaze during reading is also used as a measurement instrument in its own right, for example to compare how readers process AI-generated and human-authored text [7].

Within the Reading the Reader programme, three master’s projects built successive adaptive readers: a flexible reading and gaze-collection application [19], an adaptive e-reader used to compare four intervention presentation forms, which were found to differ substantially in disruption and acceptance [16], and a reactive frontend designed to consume machine-learning output [14]. Jensen et al. then showed that context preservation is where intervention design bites: across four intervention forms, only a gradual transition combined with highlighting significantly reduced the time readers lost to an intervention, from 14.9 s to 4.0 s [8]. Each of these systems, however, was built for its own study. Sensing, decision logic, and presentation were coupled in one application, so the next study copied and modified the previous system rather than reconfiguring a shared one. Our platform is the infrastructural complement to this line of work: it is built so that exactly such comparisons of intervention designs can be run, varied, and reproduced without touching core code.

2.2 Real-Time Gaze Processing for Reading

Reducing a raw gaze stream to oculomotor events is the classical first step, with identification algorithms catalogued by velocity, dispersion, and area-of-interest families [22] and reading behaviour extensively characterised [18]. Real-time use sharpens the problem: noisy vertical gaze must be assigned to the correct line while the reader is still reading. Kaltenberger et al. address this with confidence-scored online fixation-to-line assignment that defers uncertain assignments [9]. Our contribution on this front is architectural rather than algorithmic. The platform’s built-in analysis is a dwell-based, token-level area-of-interest method whose spatial half hit-tests gaze against the words’ live on-screen boxes (Section 3.2), so a typographic reflow cannot invalidate the mapping; and the whole analysis stage is a replaceable, logged strategy, so a stronger detector or line-assignment method can be attached as a provider without modifying the platform.

2.3 Experiment Tooling

A researcher building a reading study today would more likely reach for general-purpose experiment software than for a research prototype. Tobii Pro Lab [23], PsychoPy [15], and Psychtoolbox [5] are mature platforms for experiment design, stimulus delivery, and gaze capture. Table 1 compares them with our platform across the capabilities a gaze-driven reading experiment requires: none of the three closes the loop from live gaze to a typographic change in the text being read, and none exposes a pluggable decision boundary. Our platform occupies that unoccupied region as a focused instrument for one family of experiments; it complements rather than replaces the general-purpose tools.

3 The Platform

3.1 Overview: One Loop, Four Modules, Two Screens

The platform decomposes the adaptive reading loop into four concerns chosen along the axis of replaceability: *sensing* produces a continuous stream of gaze samples; *analysis* projects that stream into oculomotor events (fixations, saccades, regressions); *decision*

Table 1: Capability comparison of established experiment software with our platform, across the capabilities a gaze-driven reading experiment requires. Entries reflect the authors’ assessment of documented capabilities [5, 15, 23].

Capability	Tobii Pro Lab	PsychoPy	Psychtoolbox	This platform
Design experiments	Yes	Yes	Yes	Yes
Record gaze data	Yes	Partial	Partial	Yes
Analyse recorded data	Yes	Partial	No	Yes
Real-time gaze-driven text intervention	No	Partial	Partial	Yes
Pluggable decision modules	No	No	No	Yes

consumes the analysed state and returns at most a proposal to intervene; and *intervention* validates a proposal and commits a change to the text at a controlled boundary. Each concern sits behind a stable contract with a uniform shape: a module is identified by a stable provider identity and exposes a single operation that receives an immutable snapshot of the relevant session state and returns an optional result. Because a module is handed a copy of what it needs rather than a reference to live state, it can neither reach into nor mutate the core; and because the runtime resolves modules from a per-concern registry by identifier, never naming a concrete implementation, adding a module is one registration and no existing code changes. The backend (.NET) owns all authoritative session state; the dependency rule that the core references no infrastructure is enforced by the build itself, so a violation is a compile error rather than a convention.

Two browser surfaces of a single web frontend face the two people in the room (Figure 2). The participant surface renders the reading material as reflowable, tokenised text. The researcher console mirrors the same text and overlays the participant’s live gaze, a fixation heat map, and recent saccades, so the researcher sees not merely where the participant looks but how they read. The surfaces never communicate with each other; both render the same backend-owned snapshot, so they cannot disagree. Discrete commands (configure a session, change an intervention policy, start, stop) travel over a request-and-response channel, while the high-frequency gaze stream travels over a separate realtime channel (a single WebSocket per client), a split that keeps the per-sample path free of the locking that guards session lifecycle. Figure 3 traces one cycle of the loop across these parts.

3.2 Sensing and Gaze-to-Word Mapping

Sensing is a port with interchangeable sources. The primary adapter subscribes to a Tobii eye tracker’s native callback and republishes each sample (per-eye position on the display area normalised to the unit square, pupil data, validity flags, device timestamp) into the domain; a simulated source emits samples of exactly the same shape from the mouse cursor, so the full loop runs with or without hardware and the downstream stages cannot tell which source produced a sample. On the backend, the gaze path is deliberately

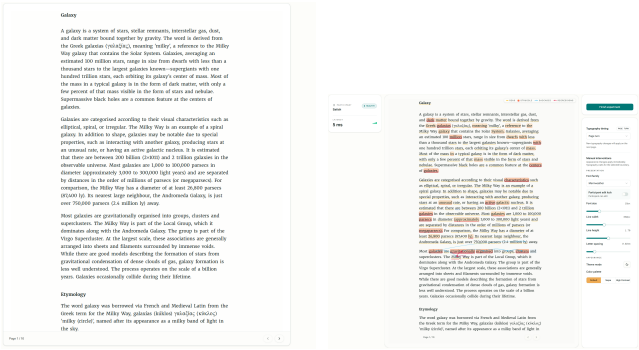


Figure 2: The two operated surfaces: the participant reader (left) and the researcher console mirroring the same text with live gaze, fixation, and saccade overlays and the intervention controls (right).

lock-free: each arriving sample is sanitised, published with a volatile write as the latest value, counted atomically, and broadcast, without acquiring the lock that serialises lifecycle commands. The heavier modular pipeline of analysis, decision, and intervention runs at the cadence of analysed-state changes rather than once per raw sample, which keeps the indirection introduced for modularity off the per-sample path. On the participant surface, snapshot bursts are coalesced to a single React commit per animation frame, so the live view re-renders at display rate regardless of the inbound message rate while never dropping the newest state.

The spatial half of the pipeline runs in the reader, the only place the rendered layout exists. Every word is a tokenised span whose on-screen box is measured from the live DOM and re-measured whenever the layout changes, prompted by mutation, resize, and scroll observers. An incoming gaze point is smoothed adaptively, mapped into the reading area, and hit-tested in two stages: first the line, scored by vertical distance with a fixed hysteresis bonus for the line currently being read, so that vertical jitter of the order of the line spacing does not flicker attribution between neighbours; then the nearest word on that line. The mapping has two properties that matter for an adaptive reader. It is resolution-independent, because the tracker reports gaze as a fraction of the display and the words are measured live rather than assumed. And it is reflow-proof by construction: a typographic intervention that moves every word to a new position cannot invalidate the mapping, because the next sample is hit-tested against the boxes as they now are, which is precisely the failure mode a coordinate-based scheme would suffer. The result of the stage is a token-level reading observation (token, line, and block identity) published back to the backend.

3.3 Eye-Movement Analysis as a Replaceable Strategy

The temporal half of the pipeline is an explicit analysis strategy behind its own contract. The built-in strategy realises the area-of-interest branch of the classical taxonomy [22]: since each observation is already bound to a word, the word itself is the area of interest, and a fixation is confirmed from accumulated dwell, with thresholds of 90 ms initially, 70 ms for a candidate on the same line,

and 135 ms across a line change, the latter raised because cross-line movement is larger and noisier. For per-word attention statistics, dwell below 45 ms is discarded as noise and dwell of at least 130 ms counts as a fixation, with the band between recorded as a skim, so the statistics separate words that were read from words the gaze merely crossed. Saccades are classified from token and line deltas, and a backward or line-change-backward saccade is counted as a regression, the standard oculomotor sign of reading disruption [18]. The strategy is a pure function of the observation and the prior state, so replaying the same observations yields the same events, and it is resolved from a registry, so an external analysis provider can replace it at runtime (Section 3.4).

3.4 The Decision Boundary and Execution Modes

Decisions about when and how to adapt the text can come from three sources: the researcher acting manually from the console, a built-in rule-based strategy, or an external provider. Two execution modes cover the supervision spectrum without structural change: in *advisory* mode every machine proposal is surfaced on the console and applied only when the researcher approves it, and in *autonomous* mode a permitted proposal is applied directly. Every proposal passes through an explicit lifecycle with exactly one terminal outcome (approved, rejected, applied, or superseded when a newer proposal or a mode change overtakes it), so the text never adapts on a decision that no longer holds, and every outcome is logged.

External providers connect out of process over a documented WebSocket protocol: a handshake in which the provider declares its identity, modules, and capabilities and receives an accept-or-reject reply, followed by heartbeats and tunnelled traffic. What crosses the boundary is a fixed vocabulary: outward, an immutable session snapshot and the stream of analysed signals (reading focus, attention summary, oculomotor events); inward, a proposal pairing an observed signal and a short human-readable rationale with the identity of an intervention module and the parameters to apply. The seam commits the platform to signals in and a proposal out, but not to any particular way of arriving at the proposal, which is what makes an AI-driven provider a configuration rather than a rewrite. An external decision is applied through exactly the same validation and logging path as a built-in one, and a provider that stops sending heartbeats or disconnects is dropped, with the affected concern degrading to its built-in fallback and the transition recorded in the session log. The same provider framework serves the analysis seam in the same way, one mechanism for two concerns.

3.5 Context-Preserving Interventions

Typographic interventions are the platform’s actuator: built-in modules cover font family, font size, line width, line height, letter spacing, theme, and palette. Appearance-only changes commit immediately. Layout-changing interventions instead wait for a chosen commit boundary (immediately, at the end of the sentence or paragraph being read, or at the page turn), so a reflow lands at a natural break rather than mid-line.

Around the commit, the reader runs a capture-and-restore mechanism. An anchor is captured continuously (every 120 ms and on

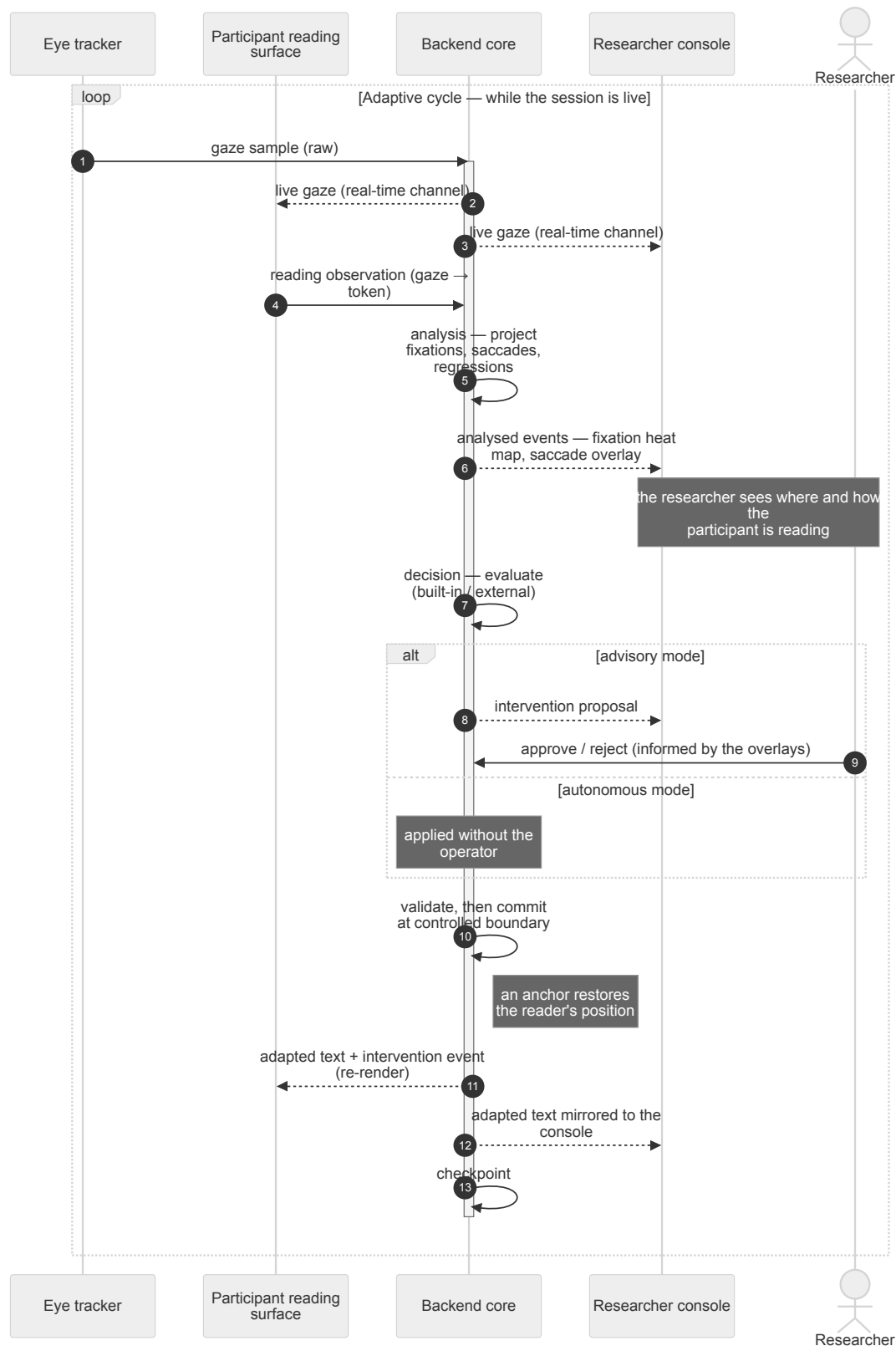


Figure 3: One iteration of the real-time adaptive loop. A gaze sample is mirrored to both surfaces over the realtime channel; the participant surface, which alone knows the rendered layout, maps it to a token and returns an enriched observation; analysis projects observations into oculomotor events; a decision (built-in, external, or the researcher acting manually) yields at most one proposal, which in advisory mode waits for researcher approval; the intervention commits at a controlled boundary and the adapted text returns to both surfaces. Persistence checkpoints are written best-effort throughout.

content changes), preferring the word the participant is currently reading, degrading to a visible fallback word or its enclosing block, and recording the element’s identity together with its viewport offset; anchors older than 15 s are rejected as stale. When a layout change commits, the reader re-locates the anchor in the reflowed document (degrading sentence to token to block) and scrolls it back to the viewport offset it held before the change. The residual error between target and achieved offset is then measured, graded (preserved, degraded, or failed, with tolerances of 72 px for a full reflow and 48 px to 120 px otherwise), and written into the session record next to the intervention that caused it. Optionally, the restored sentence is briefly highlighted and fades over about three seconds, drawing the eye back to the resume point, which corresponds to the highlighting component that prior work found effective for recovery [8].

This restore-to-captured-offset design is itself a product of the platform’s self-measurement. Our original strategy re-established the committed sentence near the top of the reading area; Section 4.4 shows how the recorded outcomes exposed that this over-repositioned the reader on small reflows, and how the revision removes the defect. Because every intervention’s graded outcome and residual are in the record, a later analysis can ask not only whether an intervention helped a reader but what it cost them in lost position.

3.6 Researcher Console, Session Record, and Replay

A session is operated end to end from the researcher console. Setup is a gated stepper: a session cannot start until a participant is saved, a licensed eye tracker is selected, a calibration has passed validation, and reading material is chosen, with the gates re-checked by the backend at the moment of start. Calibration uses 9-, 13-, or 16-point presets followed by a five-target validation that computes angular accuracy and precision, passing only when the averages stay within fixed bounds and at least four fifths of the points pass. During the run, the console shows the mirror with live analysis overlays, the current gaze-validity rate, and the measured client latency, and carries the intervention and typography controls together with the approval queue in advisory mode.

Everything that happens flows into one authoritative, schema-versioned session record owned by the backend: the run parameters and presentation baseline, calibration metrics, screen geometry, the raw gaze stream, derived oculomotor events, decision proposals and their outcomes, interventions and their context-preservation outcomes, and quiz responses, each entry stamped with a monotonic sequence number and timestamp. The record is not assembled after the fact; it is the same structure a background worker checkpoints throughout the session, so durability and export are one artifact and an interrupted session is recoverable from its latest checkpoint. A completed record exports to JSON and CSV, re-imports through a version-gated schema check, and replays in the reading interface with scrubbing, variable speed, and jump-to-event navigation. Replay is the operational definition of the record’s completeness: if a session can be re-presented and re-analysed from the record alone, the record contains what reproducibility requires. The platform also

Table 2: The eight recorded sessions. Durations and counts are taken from the exported session records.

	Condition	Dur. (s)	Samples	Fixations	Interv.
P1	With preservation	186	16664	656	5
P2	With preservation	252	22688	1525	7
P3	With preservation	171	15305	866	3
P4	With preservation	224	20136	943	4
P1	Without	475	42683	1914	9
P2	Without	123	11063	719	4
P3	Without	386	34624	1994	11
P4	Without	141	12697	694	2

emits a telemetry record (round-trip times per client, achieved sampling rate, validity, and decision-path latencies) against an explicit 100 ms budget, which is the instrumentation behind Section 4.2.

4 Evaluation

The evaluation asks four questions that mirror the platform’s claims: whether the four concerns are genuinely replaceable without touching the core (Section 4.3), whether a real eye-tracking stream drives the loop within the latency budget (Section 4.2), whether interventions can adapt the text while preserving the reader’s context (Section 4.4), and whether the researcher has the control and the auditable record needed to operate and reproduce experiments (Section 4.5). The scope is architecture and runtime feasibility, not human efficacy: with four participants, the behavioural numbers below are descriptive, and we draw no statistical inference from them.

4.1 Setup

We recorded eight reading sessions on real hardware: four participants each read an expository Markdown article under two conditions, with and without the context-preserving restore (Table 2). Sessions ran on a Tobii IS4_Large_Peripheral tracker under a 2560×1440 display, each gated by a passing nine-point calibration validation (average accuracy 0.30 to 0.69 degrees, precision 0.08 to 0.33 degrees). The deployment was the intended single-operator setting: backend and both surfaces on one host, traffic over the local loopback. All eight sessions were operated in manual mode from the researcher console, and their eye-movement analysis was served live by the external provider of Section 4.3. One additional session under an advisory decision strategy exercised the decision-path instrumentation and is reported separately in Section 4.2. Every number, figure, and table in this section is computed by a scripted, re-runnable pipeline from the exported session and telemetry records alone.

4.2 Real-Time Feasibility

Across the eight sessions the tracker delivered a stable stream at its rated rate: the achieved sampling rate, computed from device timestamps, held at a mean of 89.9 Hz (SD 0.2 Hz), with the inter-sample interval concentrated at the nominal 11.1 ms period. The gaze-validity rate, the fraction of samples with at least one eye reported valid, ranged from 92.5% to 98.0% (mean 95.6%) under

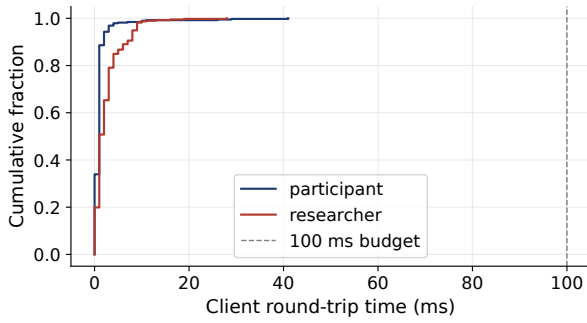


Figure 4: Cumulative distribution of client round-trip time for the participant and researcher clients against the 100 ms budget. Every sample falls within budget.

continuous, self-paced reading rather than a constrained fixation task.

The latency budget is 100 ms from a gaze sample to the dispatch of an intervention, the threshold below which an interface response is perceived as instantaneous [11, 12] and well within the duration of a single reading fixation [18]. The telemetry record carries the round-trip time between each client and the backend: the participant client saw a median of 1 ms with a 95th percentile of 3 ms, the researcher client a median of 1 ms with a 95th percentile of 8 ms, and no sample in either stream exceeded the budget (Figure 4). These are loopback transport figures of the single-host deployment, which is the intended operating point.

The decision path is instrumented separately and is recorded only when an automated strategy is active, which the eight manual-mode sessions were not. In the additional advisory-mode session (about 91 s, with the reference external decision provider running out of process and every proposal gated by the researcher), the in-process decision pipeline, timed from the freshest gaze sample to proposal dispatch, had a median latency of 6 ms, a 95th percentile of 11 ms, and a maximum of 46 ms across 14015 samples, while the round trip to the provider had a median of 5 ms and a maximum of 7 ms across its eight proposals. Both legs stayed within budget, but we read this as a validation of the instrumentation under a rule-based reference provider on one host, not as a campaign-scale result.

4.3 Modularity and External Providers

The dependency rule that makes the modules replaceable is enforced twice: the build itself admits no reference from the core outward, and an executable boundary test inspects the loaded assemblies and fails if the domain or application core depends on infrastructure, transport, or host code. Adding an in-process intervention module requires one registration entry and edits to no existing module, because the runtime resolves modules from registries by identifier.

The stronger test is out of process. Three external providers connect to the running platform over the documented provider protocol: a reference decision provider and a reference analysis provider written by us, and a connector that bridges the reading-difficulty pipeline of a concurrent, independent master’s project at

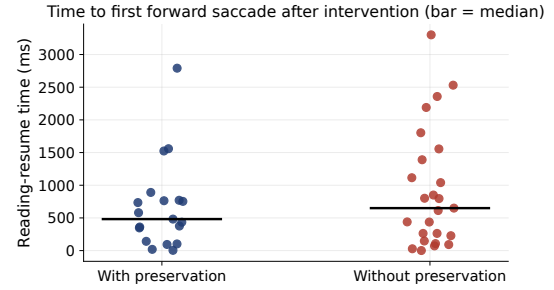


Figure 5: Reading-resume time per intervention by condition (19 interventions with preservation, 26 without; bars mark medians). Descriptive only.

our institution to the analysis seam.¹ The reference providers, written by the contract’s own authors, evidence only that the contract is sufficient; the third goes further, because we neither wrote its pipeline nor designed the contract around it, its addition changed nothing in the backend, and, most tellingly, it served as the live analysis source for the eight evaluation sessions themselves. The fixation, saccade, and regression events reported throughout this section were produced by that independent team’s code running behind the platform’s contract.

4.4 Context Preservation

The eight sessions form a with-and-without comparison of the context-preserving restore. In the enabled condition the restore ran together with its highlight cue, so the conditions contrast a bundle (positional restore plus a brief highlight of the resume point) against neither; we return to this confound below. Because preservation acts in the moments after a reflow, both behavioural measures are aligned to each intervention rather than averaged over sessions, which leaves the conditions with unequal counts: 19 interventions with preservation and 26 without.

Reading-resume time, the interval from an intervention to the reader’s next forward saccade, was shorter with preservation (median 482 ms, mean 670 ms) than without (median 650 ms, mean 923 ms), as shown in Figure 5. The regression rate in the five seconds after an intervention showed the same separation: 23% with preservation, down from a 29% pre-intervention baseline, against 33% without, essentially unchanged from its 32% baseline (Figure 6). The pattern held across three-, five-, and ten-second windows, and the resume-time direction held for three of the four participants. We state the direction of these numbers without inferring from them: four participants and an unevenly split sample do not support inference, and the bundled highlight means the difference cannot be attributed to the restore geometry alone, the more so as prior work found highlighting to be the component driving recovery improvement [8].

¹*Reading the Struggle*, by Katarina Kraljević and Sree Keerthi Desu (DTU). Their pipeline derives reader-struggle signals (cognitive load, reading style, and a pupillary load index) from eye-tracking features; our connector adapts it to the analysis contract and uses their code unmodified. [TODO: Check whether their thesis is now submitted and citable, and cite it here if so.]

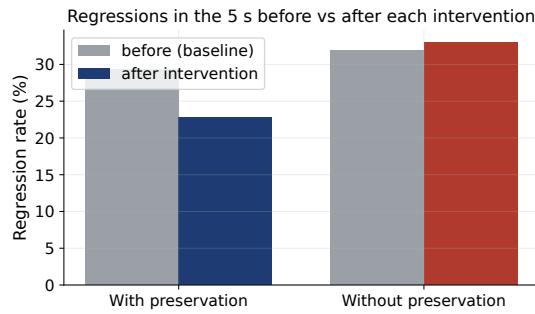


Figure 6: Regression rate in the five seconds before and after each intervention, by condition. Without preservation the rate rises after the intervention; with preservation it falls below its baseline. Descriptive only.

Self-measurement catches a defect. Within the four preservation-enabled sessions, the reader recorded a graded outcome for every intervention: the displacement the reflow induced and the residual after the restore. Sixteen of the nineteen outcomes were graded degraded, and for those the median residual was 389 px against a median induced displacement of only 38 px. The restore was moving readers substantially further than the reflow it compensated. The cause was the original strategy’s fixed target: it re-established the committed sentence near the top of the reading area regardless of how far the reader actually sat from it, so the smaller the reflow, the worse the compensation.

A deterministic sweep, and the revision. The live outcomes are observational, covering whatever interventions the participants happened to receive. To measure displacement under controlled conditions, a measurement harness mounts the real participant reader outside a live session and drives the same restore hook over every built-in layout intervention from six reading positions, with preservation off and on: 66 trials per condition. Table 3 reports the median on-screen displacement of the word the reader was on (one reading line is 32.4 px). Without preservation the median displacement is 34.3 px, though heavy-tailed (a six-point font enlargement moves the word 142 px, and line-width changes, which repaginate, up to 188 px). With the original restore, preservation made displacement worse rather than better: the median tripled to 104.4 px, the restore moved the word further than the bare reflow in 45 of 66 trials, and the effect was starkest for the smallest changes (a letter-spacing step that moves the word not at all was thrown 115 px). The revised restore returns the captured reading position to the viewport offset it held before the change (Section 3.5). It removes the over-repositioning (4 of 66 trials) and brings the median displacement to 32.4 px, at or below the no-preservation baseline, with the gains concentrated exactly where reflows are large; line-width changes repaginate and lie beyond what a within-page restore can compensate. Figure 7 shows both restore versions against the line of equality. The revision is verified geometrically by this sweep; its behavioural benefit to readers has not yet been re-measured.

Table 3: Median on-screen displacement (px) of the word being read, per intervention, from the 66-trial deterministic sweep: without preservation, with the original (top-of-area) restore, and with the revised (offset-preserving) restore. Lower is better; one reading line is 32.4 px.

Intervention	n	Without	Original	Revised
Font size +2 px	6	24.5	168.3	24.5
Font size +6 px	6	142.4	225.2	43.0
Font size −2 px	6	38.0	82.7	54.0
Line height +0.3	6	35.4	124.7	35.4
Line height −0.3	6	34.5	53.7	31.7
Line width −120 px	6	188.2	156.2	188.2
Line width +80 px	6	161.9	148.6	148.6
Letter spacing +0.06 em	6	0.0	115.0	0.0
Letter spacing +0.12 em	6	32.4	154.8	16.3
Font family → Inter	6	1.0	31.3	1.0
Font family → Space Grotesk	6	0.0	49.5	0.0
Overall	66	34.3	104.4	32.4

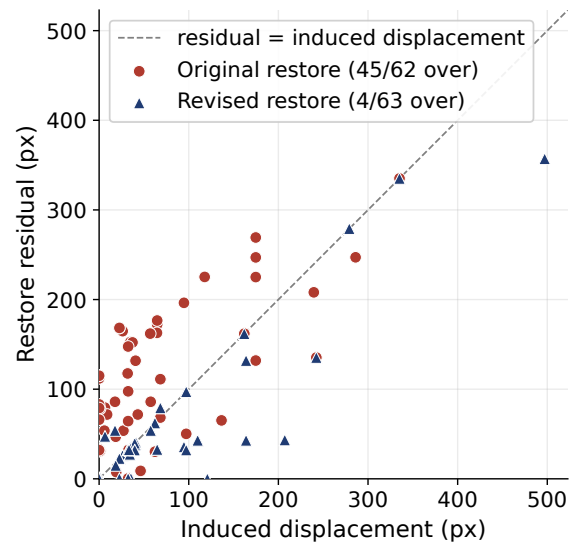


Figure 7: Restore residual against the displacement the reflow induced, one point per sweep trial for which the reading word stayed on screen in both conditions. The original restore (circles) lies above the line of equality for most trials (45 over-repositions); the revised restore (triangles) returns onto or below it (4).

4.5 Researcher Control and Reproducibility

All recorded sessions were operated end to end through the gated researcher console: no session started before its readiness gates and calibration validation passed, every intervention in Table 2 was applied from the console during the run, and the participant’s view was mirrored throughout. Reproducibility is demonstrated by construction: every figure and table in this section was produced from the exported session and telemetry records alone, with no reference

to live state, and the replay view reconstructs any recorded session from its export.

5 Discussion

What the platform enables. The comparisons this research area needs are now configuration rather than reimplementations. Contrasting intervention presentation forms, commit boundaries, or decision strategies under identical sensing is a matter of session templates and module selection, with counterbalancing supplied by the platform’s randomised item ordering and evidence supplied by the session record. The unbundled study that our own confound calls for, crossing the revised restore with and without the highlight cue, is exactly such a configuration. Beyond single studies, the exported records form a corpus on which an external decision provider can be trained offline and then attached through the same seam it will be judged at, closing the loop the programme’s envisaged AI-driven adaptation requires [6].

Measure the quality attribute you claim. The over-repositioning result is the strongest argument for a design principle the platform embodies: a system should measure the property it claims to provide, so it can report its failures as readily as its successes. Context preservation was a headline feature, yet its original geometry made small reflows worse, and nothing in ordinary operation would have surfaced that; the per-intervention residual in the session record did. The same principle governs the module boundary, whose out-of-process cost is not asserted but recorded per proposal on a single clock. We suggest this transfers beyond reading platforms: any interface that changes state during a continuous user task can hold the change until a natural boundary, re-anchor the user’s focus across it, and record what the change cost.

Limitations. The behavioural comparison rests on four participants, drawn from the authors and acquaintances, reading one article under manual intervention; the numbers are descriptive, and the enabled condition bundles the restore with the highlight cue, so no causal attribution to the restore geometry is possible. The revised restore is verified geometrically over the deterministic sweep but has not been re-evaluated with readers, so its behavioural benefit is unconfirmed. The automated decision path was exercised within budget in a single advisory-mode run with a rule-based reference provider on one host, not at campaign scale or over a network, and no learned decision model has yet run against the platform. The provider contract has one genuinely external implementation; a single outside integration is encouraging, not proof of independence. Latency figures are loopback figures of the intended single-host deployment. Reading material is restricted to reflowable Markdown, and our participants were general readers, not the vision-impaired or dyslexic readers who motivate the programme, so no claim here extends to those groups.

Future work. Three studies follow directly. First, the three-condition study separating restore geometry from highlighting on the revised mechanism, which would also replace the authors-as-participants sample. Second, a campaign-scale run of the automated decision path with a learned provider, turning the single-run latency check into a measured result. Third, the properly powered efficacy study the platform was built to host, asking whether specific interventions

improve reading rather than only measuring their immediate cost. On the platform side, additional external providers and webcam or multimodal sources behind the existing sensing port would push the contract from author-validated toward independently validated.

6 Conclusion

We asked whether the gaze-adaptive reading loop can be cleanly separated into replaceable modules under real-time conditions, and answered with a built and measured platform: a two-screen, researcher-operated system whose sensing, analysis, decision, and intervention modules sit behind contracts enforced by the build, driven by a real 90 Hz eye tracker within a 100 ms budget, validated by an independent team’s pipeline serving as its live analysis source, and recorded into replayable sessions from which every result in this paper was reproduced. Its self-measurement exposed, and its revision removed, an over-repositioning defect in our own context-preserving restore, which is precisely the kind of finding an instrument should produce about itself. The controlled studies this field needs next, unbundling restore geometry from highlighting, running learned decision providers at scale, and measuring efficacy for the readers who motivate the work, are configuration choices on this platform rather than rewrites of core code.

Ethics Statement

[TODO: Authors: state the consent arrangement for the eight recorded sessions (volunteer participants, verbal or written consent, what was stored) and the data handling (all gaze data stored locally on the recording machine, no transmission to external services, exports anonymised to participant codes before any release). This must be written by the authors from what actually happened; do not publish without it.]

Availability

The platform (backend, both reading surfaces, and the reference providers), its documentation site, the deterministic displacement harness, and the scripted analysis pipeline that produces every figure and table in this paper are open source under the MIT licence. **[TODO: Authors: decide the citable artifact link (repository plus documentation site), anonymise the committed session exports (file names currently carry participant first names), and add an anonymised link if the venue requires blinded review.]**

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